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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
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Course Outcome	COURSE OUTCOME - 1

Case Study Topic: Flipkart product catalog — price-range scans dominate.

Work Done Summary

This case study analyzed the use of a **B+ Tree index** for managing a large smartphone product catalog indexed by **(category, price)**. Since most searches are **price-range queries**, the B+ Tree efficiently retrieves products by scanning linked leaf nodes with minimal disk access. With **10^7 products** and about **800 results per query**, the structure provides fast and scalable performance. The study also considered using a **hash index** for exact lookups, but concluded that the B+ Tree is the better overall choice because it effectively supports both range searches and direct lookups.

Implementation Details

1. Implemented a **B+ Tree** index using the composite key **(category, price)**.
2. Stored **200 product records per leaf node** in an **8 KB page**.
3. Connected leaf nodes using **linked pointers** for efficient sequential access.
4. Used the B+ Tree to handle **price-range queries** quickly and efficiently.
5. Maintained a **balanced tree structure**, ensuring **$O(\log n)$** search performance.
6. Reduced **disk I/O operations** by scanning only relevant leaf pages during queries.
7. Compared with a **hash index** and selected the **B+ Tree** because it supports both exact-match and range queries effectively.

Output

Total Products: 10000000

Query

Category : Electronics

Price Range : [12000, 14800]

Results

Products Found : 800

Pages Scanned : 4

Range Scan : Successful

Workload Analysis

Range Queries : 87%

Exact Lookups : 5%

Conclusion

B+ Tree efficiently handles range queries

Minimal disk I/O operations

B+ Tree preferred over Hash Index

Result

1. Successfully retrieved products within the specified price range using the **B+ Tree index**.
2. Found approximately **800 matching products** with only **4 page scans**.
3. Achieved **low disk I/O cost** and efficient range-query performance.
4. Demonstrated that the **B+ Tree** is highly suitable for large-scale product catalogs.
5. Concluded that the **B+ Tree outperforms a hash index for range-based searches** while maintaining efficient lookup operations.

Time Complexity

Operation	B+ Tree
Insert	$O(\log n)$
Search	$O(\log n)$
Delete	$O(\log n)$
Range Query	$O(\log n + k)$
Exact Lookup	$O(\log n)$

Conclusion

The B+ Tree proved to be an efficient indexing structure for managing a large product catalog containing 10^7 products. It supports fast search, insertion, and range-query operations with $O(\log n)$ complexity while reducing disk I/O through linked leaf nodes. Since most user queries are based on price ranges, the B+ Tree performs more effectively than a hash index for this workload. Therefore, it is a suitable choice for scalable and efficient product retrieval in large e-commerce platforms.

GitHub Link: <https://github.com/vigneshkumar-stack/DSA-Project>

Student Signature	Faculty Signature